

Numerical Periods and Quasi-Periods of Δ and $\widehat{\Delta}$ via Finite-Endpoint Cocycles

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Abstract

We reproduce Brown's numerical values for the periods $\omega^\pm$ of the modular discriminant Δ and the quasi-periods $\eta^\pm$ of the weakly holomorphic cusp form $\widehat{\Delta} \in S_{12}^!$, using the finite-endpoint Eichler–Shimura cocycle formula stated by Diamantis and Rolén. The construction avoids any divergent cuspidal integrals: every contour is a finite line segment in the upper half-plane, bounded away from the cusp. We obtain agreement with Brown's reported values to all reported digits, with internal consistency at ~ 30 decimal places.

1 Setup

Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Fix the weight $k = 12$ and write $n = k - 2 = 10$.

Let $V_n = \bigoplus_{i+j=n} \mathbb{Q} X^i Y^j$ with the right Γ -action $(X, Y)|_\gamma = (aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. A 1-cocycle is a map $C : \Gamma \rightarrow V_n$ satisfying $C_{gh} = C_g|_h + C_h$, uniquely determined by (C_S, C_T) .

We work with two modular forms:

- $\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24}$, the unique normalized cusp form in S_{12} .
- $\widehat{\Delta} \in S_{12}^!$, the unique weakly holomorphic cusp form with q -expansion

$$\widehat{\Delta} = q^{-1} + 0 + 0 \cdot q + 47709536 q^2 + 39862705122 q^3 + 7552626810624 q^4 + \cdots \quad (1)$$

2 Construction of $\widehat{\Delta}$ via $\Delta \cdot J$ -products

Let $j(\tau) = E_4^3/\Delta = q^{-1} + 744 + 196884 q + \cdots$ be the modular j -invariant, and set

$$J := j - 744 = q^{-1} + 0 + 196884 q + 21493760 q^2 + \cdots$$

the normalized Hauptmodul. Since Δ has a simple zero at the cusp and J^k has a pole of order k , the product $\Delta \cdot J^k$ is weakly holomorphic of weight 12 with leading q -behaviour q^{1-k} . We seek $\widehat{\Delta}$ in the form

$$\widehat{\Delta} = \Delta \cdot (J^2 + a_1 J + a_0) \quad (2)$$

with $a_1, a_0 \in \mathbb{Q}$ to be determined. The q^{-1} Fourier coefficient of $\Delta \cdot J^2$ is 1 automatically (from $q \cdot q^{-2}$), and the q^{-1} coefficients of $\Delta \cdot J$ and Δ vanish; so the q^{-1} condition in (1) is automatic. The q^0 and q^1 conditions yield a 2×2 linear system whose solution is

$$\boxed{a_1 = 24, \quad a_0 = -393444.} \quad (3)$$

Computing $\widehat{\Delta} = \Delta(J^2 + 24J - 393444)$ to order q^4 exactly reproduces Brown's reported values 47709536, 39862705122, and 7552626810624.

3 The Diamantis–Rolen finite-endpoint cocycle

Fix any $\tau_0 \in \mathbb{H}$. For any cusp form (weakly holomorphic or otherwise) f of weight k , define

$$\sigma_f(\gamma)(z) := \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (\tau - z)^{k-2} d\tau, \quad \gamma \in \Gamma. \quad (4)$$

This is the bottom formula in Diamantis–Rolen [2, §2.1], valid for cusp forms in the (broad) sense that the resulting cohomology class is cuspidal. Crucially, the contour is a finite segment and stays inside $\mathbb{H}$; no regularization is required, even when f has a pole at the cusp. Changing τ_0 alters σ_f only by a coboundary, so the class $[\sigma_f] \in H^1(\Gamma; V_n)$ is well-defined.

Brown’s equivalent definition [1, Def. 2.4, eq. (2.12)] is

$$C_f^\gamma(X, Y) = F_f(\gamma\tau)|_\gamma - F_f(\tau), \quad F_f(\tau) := \int_{\tau_0}^{\tau} (2\pi i)^{n+1} f(w) (X - wY)^n dw, \quad (5)$$

which one shows to be independent of τ by direct computation, and to coincide (after the cosmetic substitution $z \leftrightarrow X/Y$ and the prefactor $(2\pi i)^{n+1}$ with (4).¹ We absorb the $(2\pi i)^{n+1}$ factor into our cocycle below to match Brown’s normalization.

4 Procedure

For $f \in \{\Delta, \hat{\Delta}\}$, fix $\tau_0 \in \mathbb{H}$ generic. Define the cocycle values

$$C_S := (2\pi i)^{11} \int_{-1/\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^{10} d\tau, \quad C_T := (2\pi i)^{11} \int_{\tau_0-1}^{\tau_0} f(\tau) (X - \tau Y)^{10} d\tau.$$

Both are polynomials in $V_{10} \otimes \mathbb{C}$.

Cuspidality. A cocycle is *cuspidal* iff $C_T = 0$. For Δ , taking $\tau_0 \rightarrow i\infty$ formally would give $C_T = 0$ identically; with finite τ_0 , $C_T \neq 0$ but its cohomology class is cuspidal. The same is true for $\hat{\Delta}$. The cuspidality manifests numerically as

$$[C_T]_{X^{10}} = 0,$$

since the X^{10} direction is the unique T -invariant in V_{10} .

Coboundary modification. We seek $P \in V_{10}$ such that $P|_T - P = C_T$, so that the modified cocycle $C'_\gamma := C_\gamma - (P|_\gamma - P)$ satisfies $C'_T = 0$. Writing $P = \sum_{j=0}^{10} p_j X^{10-j} Y^j$, the relation $P|_T(X, Y) = P(X + Y, Y)$ produces an upper-triangular linear system: the equation for the $X^{10-m} Y^m$ coefficient ($m \geq 1$) reads

$$\sum_{j=0}^{m-1} p_j \binom{10-j}{m-j} = [C_T]_{X^{10-m} Y^m},$$

¹The check is straightforward: in the difference $F_f(\gamma\tau)|_\gamma - F_f(\tau)$, the variable τ contributions cancel via the modularity of $f(w)(X - wY)^n dw$ under the slash action, leaving the basepoint-only piece $\int_{\gamma^{-1}\tau_0}^{\tau_0} (2\pi i)^{n+1}$.

which determines $p_0, \dots, p_9$ recursively, while the X^{10} equation reads $0 = [C_T]_{X^{10}}$ — exactly the cuspidality condition. The coefficient p_{10} does not appear in any equation: Y^{10} is T -invariant ($T \cdot Y = Y$), so it lies in $\ker((\cdot)|_T - (\cdot))$.

The freedom $p_{10} \in \mathbb{C}$ is therefore real but inconsequential. Replacing $P \rightarrow P + cY^{10}$ shifts C'_S by

$$c \cdot (Y^{10}|_S - Y^{10}) = c \cdot ((-X)^{10} - Y^{10}) = c(X^{10} - Y^{10}),$$

which is itself a coboundary in V_{10} — i.e., it shifts C'_S within its cohomology class. So the periods $\omega^\pm, \eta^\pm$ defined cohomologically are independent of p_{10} . We fix $p_{10} = 0$ by convention. The resulting C'_S has unambiguous coefficients on every monomial except X^{10} and Y^{10} (which carry the p_{10} -dependent shift); the interior monomials $X^{10-j}Y^j$ for $1 \leq j \leq 9$ are unaffected by this choice.

In particular,

$$C'_S = C_S - (P|_S - P), \quad P|_S(X, Y) = P(Y, -X).$$

Period extraction. Brown's basis polynomials of $H^1_{\text{cusp}}(\Gamma; V_{10})$ [1, §8.2] are

$$\begin{aligned} P_S^+ &= \frac{36}{691}(Y^{10} - X^{10}) + X^2Y^2(X^2 - Y^2)^3, \\ P_S^- &= 4X^9Y - 25X^7Y^3 + 42X^5Y^5 - 25X^3Y^7 + 4XY^9. \end{aligned}$$

These are values at S of the cuspidal cocycles $P^\pm$ (with $P_T^\pm = 0$); their coefficients on X^{10} and Y^{10} — including the $\frac{36}{691}$ in P_S^+ — are dictated by the cocycle equations and the $(ST)^3 = 1$ relation, *not* by any choice of coboundary representative.

The cocycle C'_S lies in a 3-dimensional affine subspace spanned by P_S^+, P_S^- , and $X^{10} - Y^{10}$:

$$C'_S = \omega^+ P_S^+ + \omega^- P_S^- + c \cdot (X^{10} - Y^{10}), \tag{6}$$

where $\omega^\pm$ are the cohomologically meaningful periods and $c \in \mathbb{C}$ depends on the choice $p_{10} = 0$ above. Reading off coefficients on *interior* monomials (those for which $X^{10} - Y^{10}$ has zero coefficient) gives $\omega^\pm$ unambiguously. Concretely:

$$\omega^- = \frac{[C'_S]_{X^9Y}}{4}, \quad \omega^+ = [C'_S]_{X^8Y^2}.$$

Internal consistency is verified by reading off from the other interior monomials: any of $X^9Y, X^7Y^3, X^5Y^5, X^3Y^7, XY^9$ for ω^- , and any of $X^8Y^2, X^6Y^4, X^4Y^6, X^2Y^8$ for ω^+ .

5 Implementation Notes

Code provenance

The numerics in Section 6 below are taken directly from the executed Jupyter notebook [3]. The notebook (a) builds the q -expansions of $\Delta, j, J = j - 744$, and $\widehat{\Delta}$ as exact Laurent series via `sympy` rationals; (b) implements the cocycle integration in `mpmath` at 40-digit precision; (c) runs the period-extraction pipeline of Section 4 for both Δ and $\widehat{\Delta}$; (d) verifies basepoint independence by re-running with a second choice of τ_0 . The full output is reproducible by running `jupyter nbconvert --execute` on the notebook.

Path geometry

We choose $\tau_0 = 0.3 + 1.2i$, giving:

- S -path: $-1/\tau_0 = -0.196\dots + 0.784\dots i \rightarrow \tau_0$, a line segment in $\mathbb{H}$.
- T -path: $\tau_0 - 1 = -0.7 + 1.2i \rightarrow \tau_0$, horizontal segment at $\text{Im}(\tau) = 1.2$.

Both paths are bounded away from $\partial\mathbb{H}$. With $\text{Im}(\tau) \geq 0.78$ throughout, $|q| = e^{-2\pi \text{Im}\tau} \leq 0.007$, so the η^{24} -product for Δ converges in ~ 200 terms at 40-digit precision. The q -expansion of $\hat{\Delta}$ has its q^{-1} pole bounded by $|q^{-1}| \leq 1882$, harmless.

Numerics

We use `mpmath` at 40 decimal digits. Each cocycle integral $\int f(\tau)\tau^j d\tau$ for $j = 0, \dots, 10$ is computed by Gauss–Legendre quadrature along the line segment; `mpmath.quad` handles this comfortably. The total runtime per form is roughly 30 seconds.

The q -expansion of $\hat{\Delta}$ is built symbolically (exact rationals via `sympy`) by computing $\Delta \cdot J^2$, $\Delta \cdot J$, Δ as Laurent series and forming the linear combination with $a_1 = 24$, $a_0 = -393444$.

6 Results

All numerical quantities below are extracted from the executed notebook [3] at 40-digit working precision and reported here to ~ 20 digits.

Δ : classical periods

	Extracted [3]	Brown [1, §8.3]
ω^+	$-68916772.80959519475431012\dots$	$-68916772.809595194754\dots$
ω^-/i	$-5585015.379310401866877139\dots$	$-5585015.3793104018668\dots$

The five odd-monomial extractions of ω^- (from X^9Y , X^7Y^3 , X^5Y^5 , X^3Y^7 , XY^9) agree pairwise to ~ 35 digits. The four interior even-monomial extractions of ω^+ (from X^8Y^2 , X^6Y^4 , X^4Y^6 , X^2Y^8) agree pairwise to ~ 33 digits.

$\hat{\Delta}$: quasi-periods

	Extracted [3]	Brown [1, §8.3]
η^+	$127202100647.17709477731716\dots$	$127202100647.17709477\dots$
η^-/i	$10276732343.64913275081719\dots$	$10276732343.649132750\dots$

Internal consistency: the five odd-monomial η^- extractions agree to ~ 28 digits; the four interior η^+ extractions agree to ~ 28 digits. (The slightly worse internal agreement for $\hat{\Delta}$ vs Δ is consistent with the larger magnitudes involved — $|\eta^\pm| \sim 10^{11}$ vs $|\omega^\pm| \sim 10^7$ — and the additional numerical work involved in evaluating the truncated q -expansion of $\hat{\Delta}$.)

Convention note

Our extraction yields ω^-, η^- as i times Brown’s reported real values. This is consistent with Brown’s period-matrix convention [1, eq. (2.13)], in which the matrix entries are

$$\mathcal{P} = \begin{pmatrix} \eta^+ & \omega^+ \\ i\eta^- & i\omega^- \end{pmatrix},$$

i.e., he absorbs a factor of i into the basis on the minus side. The magnitudes match exactly to all digits Brown reports.

Basepoint independence check

The notebook also re-runs the extraction with $\tau_0 = -0.2+0.9i$ in place of $\tau_0 = 0.3+1.2i$, reproducing η^+ and η^- to agreement at the $\sim 10^{-27}$ absolute level (i.e., ~ 38 matching significant digits given $|\eta^\pm| \sim 10^{11}$), confirming the cohomological invariance numerically.

7 Conclusion

The Diamantis–Rolen finite-endpoint formula (4) delivers the cocycle representative needed for Brown’s period-extraction recipe directly, without divergent cusp integrals or regularization. Combined with the explicit Δ -times-Hauptmodul-polynomial construction $\hat{\Delta} = \Delta(J^2 + 24J - 393444)$, the procedure reproduces all four reported periods to full reported precision. The same code, with $f = \hat{\Delta}$ swapped in for $f = \Delta$, computes (η^+, η^-) from (ω^+, ω^-) with zero structural change.

References

- [1] F. Brown, *A class of non-holomorphic modular forms III: real analytic cusp forms for $\mathrm{SL}_2(\mathbb{Z})$* , arXiv:1710.07912 (October 2017).
- [2] N. Diamantis and L. Rolen, *Period polynomials, derivatives of L -functions, and zeros of polynomials*, arXiv:1707.04814 (July 2017).
- [3] *delta_and_hatdelta_periods.ipynb* (executed Jupyter notebook), companion source file for this writeup. Implements the cocycle-based period-extraction pipeline at 40-digit `mpmath` precision; all numerical values reported in Section 6 are taken directly from its output.